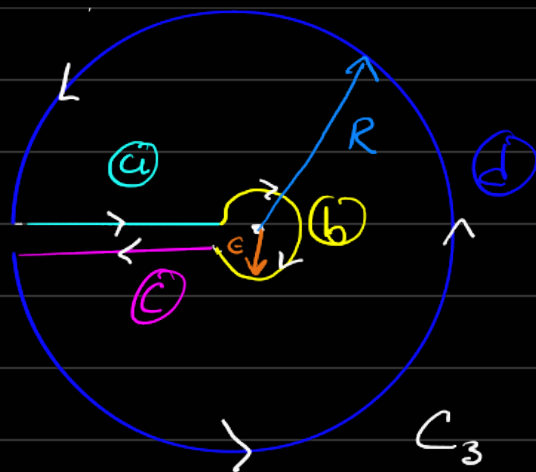


* Here we'll look at:

→ We break down each integration into the four colored regions shown



$$\oint_{C_3} = \int_{(a)} + \int_{(b)} + \int_{(c)} + \int_{(d)}$$

* Now, the idea is to take the Real integral and extend it to the complex plane via C_3 and take the limits $R \rightarrow \infty$ and $\epsilon \rightarrow 0$. Now it's important to note that the real variable has $\int_0^\infty dx : [0, \infty]$

However the contour, as constructed, goes from $-\infty$ to 0 along (a) & (c). So we need to make note of this during our integration.

Start with (3c)

(a) Here r : goes from R to ϵ : $r \in (R, \epsilon)$
 θ : $\theta = \pi$

Thus $z = r e^{i\pi} = -r$

$dz = -dr$

$$\int_{(a)} \frac{z^{1/2}}{1 + \sqrt{2}z + z^2} dz = \lim_{\substack{R \rightarrow \infty \\ \epsilon \rightarrow 0}} \int_R^\epsilon \frac{(-r)^{1/2}}{1 - \sqrt{2}r + r^2} (-dr) = -i \lim_{\substack{R \rightarrow \infty \\ \epsilon \rightarrow 0}} \int_R^\epsilon \frac{r^{1/2}}{1 - \sqrt{2}r + r^2} dr$$

we can safely take the limit:

$$= -i \int_0^\infty \frac{r^{1/2}}{1 - \sqrt{2}r + r^2} dr$$

flip integral:

$$(c) = i \int_0^\infty \frac{r^{1/2}}{1 - \sqrt{2}r + r^2} dr$$

Here $\theta = -\pi$, r goes from ϵ to R
 $z = r e^{-\pi i}$
 $dz = -dr$

$$\int \frac{z^{1/2}}{1+\sqrt{2}z+z^2} dz = \lim_{\substack{R \rightarrow \infty \\ \epsilon \rightarrow 0}} \int_{\epsilon}^R \frac{r^{1/2} e^{-\pi/2 i}}{1-\sqrt{2}r+r^2} (-dr)$$

$$\begin{aligned} &= \lim_{\substack{R \rightarrow \infty \\ \epsilon \rightarrow 0}} \int_{\epsilon}^R \frac{r^{1/2} (-i) (-1)}{1-\sqrt{2}r+r^2} dr \\ &= \lim_{\substack{R \rightarrow \infty \\ \epsilon \rightarrow 0}} i \int_{\epsilon}^R \frac{r^{1/2}}{1-\sqrt{2}r+r^2} dr \end{aligned}$$

$$= i \int_0^{\infty} \frac{r^{1/2}}{1-\sqrt{2}r+r^2} dr$$

adding (A) + (C) :

$$(A) + (C) = 2i \int_0^{\infty} \frac{r^{1/2}}{1-\sqrt{2}r+r^2} dr$$

replace: $r = x$
 $= 2i \int_0^{\infty} \frac{x^{1/2}}{1-\sqrt{2}x+x^2} dx$

which is the integral we're looking for. In general the terms along the cut and the other two terms will give zero as we will now show. Thus

$$\begin{aligned} \oint_{\gamma} \frac{z^{1/2}}{1+\sqrt{2}z+z^2} dz &= 2^{3/2} \pi i \sin\left(\frac{3\pi}{4}\right) = \left(\int_{\text{(A)}} + \int_{\text{(C)}} \right) \left(\frac{z^{1/2}}{1+\sqrt{2}z+z^2} dz \right) + \left(\int_{\text{(B)}} + \int_{\text{(D)}} \right) \left(\frac{z^{1/2}}{1+\sqrt{2}z+z^2} dz \right) \\ &= 2i \int_0^{\infty} \frac{x^{1/2}}{1-\sqrt{2}x+x^2} dx + 0 \end{aligned}$$

Thus,

$$\int_0^{\infty} \frac{x^{1/2}}{1-\sqrt{2}x+x^2} dx = \sqrt{2} \pi \sin\left(\frac{3\pi}{4}\right)$$

$$\int_0^1 \sqrt{1-x^2} dx$$

Let's now show (b) + (d) = 0

(b) Here $r = \epsilon$ and $\theta \in [\pi, -\pi]$

$$\int \frac{z^{1/2}}{1+\sqrt{2}z+z^2} dz = \lim_{\substack{R \rightarrow \infty \\ \epsilon \rightarrow 0}} \int_{-\pi}^{-\pi} \frac{\epsilon^{1/2} e^{i\theta/2}}{(1+\sqrt{2}\epsilon e^{i\theta} + \epsilon^2 e^{2i\theta})} (\epsilon i e^{i\theta} d\theta)$$

$$= \lim_{\epsilon \rightarrow 0} \int_{-\pi}^{-\pi} \left(\frac{\epsilon^{3/2} i e^{3i\theta/2}}{1+\sqrt{2}\epsilon e^{i\theta} + \epsilon^2 e^{2i\theta}} \right) d\theta$$

This goes to zero in the limit

$$\lim_{\epsilon \rightarrow 0} \left(\frac{\epsilon^{3/2} i e^{3i\theta/2}}{1+\sqrt{2}\epsilon e^{i\theta} + \epsilon^2 e^{2i\theta}} \right) = 0$$

so the integral does as well, hence

$$(b) = 0$$

(d) Here $r = R$, $\theta \in [-\pi, \pi]$
 $z = R e^{i\theta}$, $dz = R i e^{i\theta} d\theta$

$$(d) = \lim_{R \rightarrow \infty} \int_{-\pi}^{\pi} \frac{R^{1/2} e^{i\theta/2}}{1+\sqrt{2}R e^{i\theta} + R^2 e^{2i\theta}} (R i e^{i\theta} d\theta)$$

$$= \lim_{R \rightarrow \infty} \int_{-\pi}^{\pi} \frac{R^{3/2} e^{3i\theta/2} i}{1+\sqrt{2}R e^{i\theta} + R^2 e^{2i\theta}} d\theta$$

factor $R^{3/2}$ from denominator
 $\frac{1}{R^{3/2}} \left(\frac{1}{1+\sqrt{2}e^{i\theta} + e^{2i\theta}} \right) \frac{1}{R^{3/2}}$

$$\lim_{R \rightarrow \infty} \int_{-\pi}^{\pi} \frac{R^{3/2} \left[\frac{1}{R^{3/2}} + \frac{1}{R^{1/2}} e^{i\theta} + R^{1/2} e^{2i\theta} \right]}{R^{3/2}} d\theta$$

Note: $\lim_{R \rightarrow \infty} \left(\frac{1}{R^\alpha} \right) = 0, \quad \alpha > 0$

Thus
$$\int_{-\pi}^{\pi} \frac{e^{3\theta i/2} i d\theta}{(0 + 0 + \lim_{R \rightarrow \infty} (R^{1/2} e^{2i\theta}))}$$

$$= \int_{-\pi}^{\pi} \frac{e^{3\theta i/2} i d\theta}{\infty} \rightarrow 0$$

and so

$$\textcircled{1} = 0$$

as claimed.

